FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – II
PAPER –1
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABC

Sol. Total quantity of charge carried; by one pulse of carried is

$$Q = \int I dt$$

which is the area at the triangle in figure.

$$Q = \frac{1}{2} \times (20 \times 10^{-9})(80 \times 10^{-6})$$

$= 8 \times 10^{-13}$ C

The number of electrons carried by one pulse is

$$n = \frac{Q}{e} = \frac{8 \times 10^{-13}}{1.6 \times 10^{-19}} = 5 \times 10^{6}.$$

Number of photoelectrons emitted per light pulse is $n' = \frac{n}{10^5} = 5.$

Number of photons in one light pulse is $N = \frac{n'}{0.1} = 50.$

2. ACD

Sol. Time = $\frac{\text{distance}}{\text{velocity}} = \frac{S_1}{C/n_1} + \frac{S_2}{C/n_2} + ... + \frac{S_m}{C/n_m} = \frac{1}{c}\sum_{i=1}^{m} n_i s_i$

Optical path length = $n_1S_1 + n_2S_2 + \ldots\ldots + n_mS_m = \sum_{i=1} n_i s_i$

3. AC

Sol. From the relation $eV = \frac{hc}{\lambda} - \phi$

$$V = \frac{1}{e}\left(\frac{hc}{\lambda} - \phi\right)$$

The equation of straight line

slope is $\tan\theta - \frac{hc}{e}$

$$\phi_1 : \phi_2 : \phi_3 = \frac{hc}{\lambda_{01}} : \frac{hc}{\lambda_{02}} : \frac{hc}{\lambda_{03}} = 1:2:4$$

4. A

Sol. Static friction

$$f = \frac{mv^2}{R} \quad \ldots(1)$$

$f = \mu_s.N$

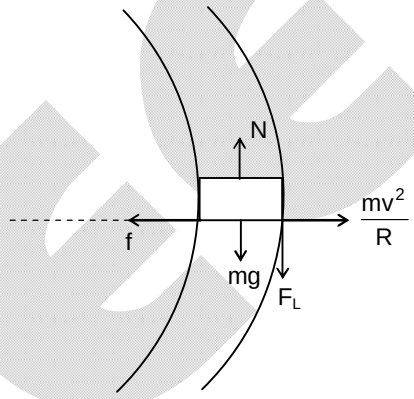
$N = mg + F_L$

F_L = lift for a acting downward

$$\mu_S N = \frac{mv^2}{R}$$

$$\mu_S(mg + F_L) = \frac{mv^2}{R}$$

$$F_L = m\left(\frac{v^2}{\mu_S R} - g\right)$$



5. A

Sol. $$\left[\frac{Q}{A}\right] = [\eta]^x \left[\frac{S\Delta Q}{h}\right]^y \left[\frac{1}{\rho g}\right]^{yz}$$

Solving weight x = 1, y = 1, z = 0.

So, $\frac{Q}{A} = \eta\frac{S\Delta\theta}{h}$

6. C

Sol. $n_Q = (341 \pm 3)$ Hz

$n_Q = 344$ Hz or $n_Q = 338$ Hz

On loading with wax, beat frequency decreases so

$N_Q = 344$ Hz.

7. C

Sol. The binding energies of the reactants and the products in the given nuclear reactions are as follows:

Reaction	Reactant	Products
(a) $Y \rightarrow 2Z$	60 × 8.5 = 510 MeV	2 × 30 × 5.0 = 300 MeV
(b) $W \rightarrow X + Z$	120 × 7.5 = 900 MeV	(900 × 8.0 + 30 × 5.0) = 870 MeV
(c) $W \rightarrow 2Y$	120 × 7.5 = 900 MeV	2 × 60 × 8.5 = 1020 MeV

(d) $X \rightarrow Y + Z$ $\quad 90 \times 8.0 = 720$ MeV $\quad (60 \times 8.5 + 30 \times 5.0) = 660$ MeV

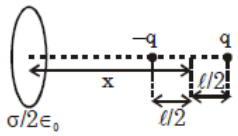
The binding energy of the products in reaction (c) is greater than that of the reactant. Hence reaction (c) release energy.

8. B

Sol. Conceptual

9. B

Sol.



(P) $E = \dfrac{\sigma}{2\in_0}$

$F = qE_- - qE_+$

$F = 0$

(Q) $F = q\left[\dfrac{\rho(x-\ell/2)}{\in_0} - \dfrac{\rho(x+\ell/2)}{\in_0}\right]$

$F = \dfrac{q\rho\ell}{\in_0}$

(R) $F = q\left[\dfrac{\lambda}{2\pi\in_0(x-\ell/2)} - \dfrac{\lambda}{2\pi\in_0(x+\ell/2)}\right]$

(S) $F = \dfrac{Q}{4\pi\in_0(x-\ell/2)^2} - \dfrac{Q}{4\pi\in_0\left(x+\dfrac{\ell}{2}\right)^2}$

10. D

Sol. $ma_y = F\cos\theta$

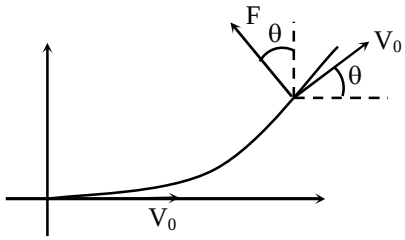
$\dfrac{mdv_y}{dt} = (B_0x)qV_0\cos\theta$

$\dfrac{mdv_y}{dx}\times\dfrac{dx}{dt} = B_0qxV_x$

$\dfrac{dV_y}{dx} = \dfrac{B_0q}{m}x$

$\Rightarrow \quad V_y = \dfrac{B_0q}{2m}x^2$

$X_{max} = \sqrt{\dfrac{2mV_0}{qB_0}}$



For radius of curvature,

$\dfrac{mv_0^2}{R} = qV_0B$

$R = \dfrac{mV_0}{qB}$

$\Rightarrow \quad R\big]_{at\ x=0} = \infty$

$R\big]_{at\ x=x_{max}} = \dfrac{mV_0}{q(B_{x=x_{max}})}$

11. C

Sol. Momentum conservation

$mv = 3mv' \ ; v' = \dfrac{v}{3}$

Energy conservation

$\dfrac{1}{2}mv^2 = \dfrac{1}{2}(3m)\left(\dfrac{V}{3}\right)^2 + \dfrac{1}{2}kx^2$

$x = \sqrt{\dfrac{2}{3}\dfrac{mv^2}{k}}$

SECTION – B

12. 2

Sol. The pressure exerted by the weight at the piston = $\dfrac{mg}{A} = 1\times10^4\ N/m^2$

Using, $P_1V_1 = P_2V_2$

$= P_1(Al_1) = P_2(Al_2)$

$\Rightarrow\ l_2 = \dfrac{P_1l_1}{P_2}$ = 220 cm.

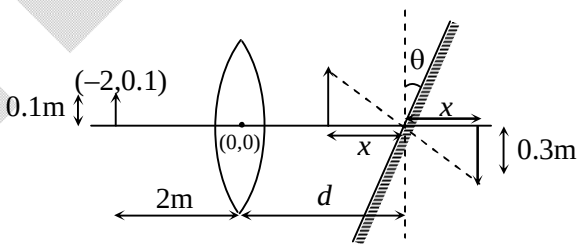
13. 5

Sol. $\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{1}{f}$

$\dfrac{1}{v} - \dfrac{1}{-2} = \dfrac{1}{1.5}$; v = 6m

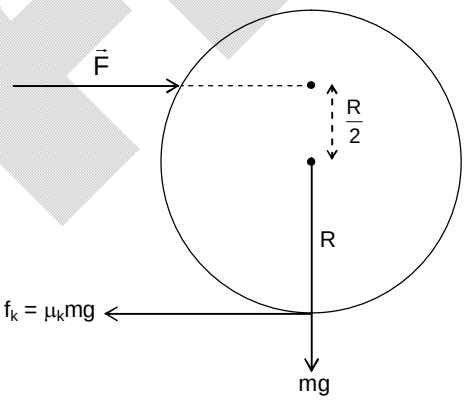
$m = \dfrac{v}{u} = -3$

$x = 6 - d$; $\tan\theta = \dfrac{0.3}{x} \quad \Rightarrow$ d = 5m.



14. 7

Sol.



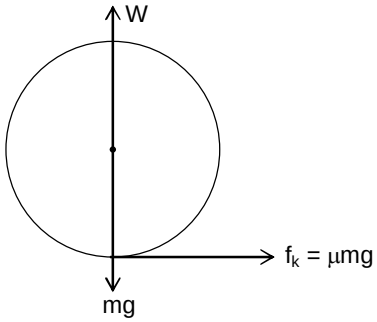
$\overrightarrow{\Delta P} = \vec{F}\Delta t = m(10-0) = 10\ m$

$\Delta\vec{L}_r$ Centre $= F\Delta t\dfrac{R}{2} = \dfrac{2}{5}mR^2\omega$

$\Delta L_r = 10\ m \times \dfrac{R}{2} = \dfrac{2}{5}mR^2\omega$

$R\omega = \dfrac{25}{2}$

$V_{cm} = 10$, $R\omega = 12.5$



$$a_{cm} = \frac{\mu mg}{m} = \mu g$$

$$\tau_{\alpha \cdot c} = \mu\, mgR = \frac{2}{5}mR^2\alpha$$

Let after time t

$$V'_{cm} = \omega' R$$

$$10 + \mu_k gt = \left[\frac{12.5}{R} - 2.5\frac{\mu_k g}{R}\right]R$$

$= 10 + \mu_k gt = 12.5 - 2.5\ \mu_k g$

$35\ \mu_k gt = 2.5$

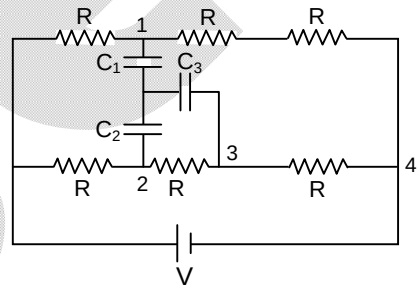
$$t = \frac{1}{7} \Rightarrow \frac{1}{t} = 7.$$

15. 18

Sol. No current passes through capacitor in steady state.

Let $V_4 = 0$

Then $V_1 = V_2 = \dfrac{2V}{3}$

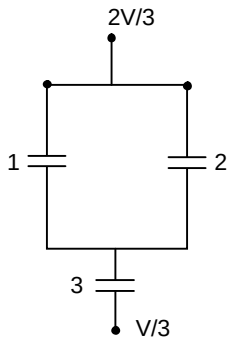


C_1 & C_2 are parallel.

Then $V_3 = \dfrac{V}{3}$

Potential difference across capacitor C_3 is

$$= \frac{2C}{2C + C} \times \left(\frac{2V}{3} - \frac{V}{3}\right) = \frac{2V}{9}$$



16. 5

Sol. $t = \pi R / v = 2u\sin\theta / g$ ....(i)

and $(u\cos\theta)t = 2R$ ....(ii)

from (i) and (ii)

u = 20 m/s

$4k = 20 \Rightarrow \quad k = 5\ m$

17. 4

Sol. $f_0 = \frac{V}{4\ell} = \frac{1}{4(\ell + x)}$

$= \frac{V}{4}\left(\frac{1}{\ell} - \frac{1}{\ell + x}\right) = \frac{V}{4}\left(\frac{t + xt}{\ell(\ell + x)}\right)$

$= \frac{Vx}{41^2}\ (x << 1)$

$= \frac{320 \times 2 \times 10^{-2}}{4 \times 2 \times 2} = 40 \times 10^{-2} = 0.4$

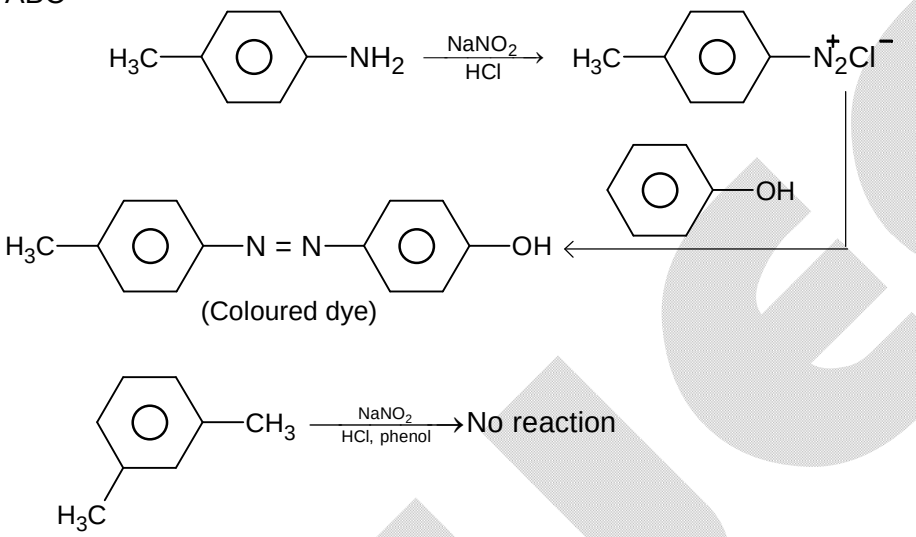
Chemistry

PART – II

SECTION – A

18. ABCD
Sol. Chromyl chloride test.

19. ABC
Sol.

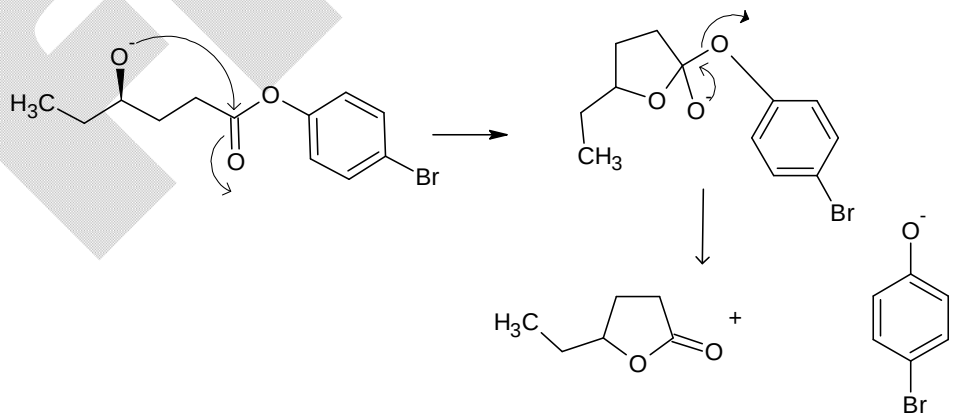


20. ACD
Sol. Azeotropic mixtures have same composition in liquid as well as in vapour phase.

21. D
Sol. $H_2(2) = \sigma(1s)^1\sigma^*(1s)^1$
Bond order = 0
So this molecule will not exist and will break down into atoms.

22. C
Sol. $PbI_4 \longrightarrow PbI_2 + I_2$ due to inert-pair effect.

23. B
Sol.



24. A
Sol. Due to electronegativity of peripheral atoms.

25. A
Sol. MnO_2 can oxidized allylic -OH into carbonyl group.

26. C
Sol. P $\rightarrow$ Nucleophilic substitution reaction followed by diazotization and rearrangement.
Q $\rightarrow$ Beckmann rearrangement
R $\rightarrow$ Baeyer's Villiger reaction
S $\rightarrow$ Pinacol pinacolone rearrangement

27. B
Sol. Raoult's law: Relative lowering of vapour pressure is equal to mole fraction of non-volatile solute.

28. A
Sol. (P) Zn $\rightarrow [Ar]_{18}4s^2 3d^{10}$. In +2 O.S, it contains no unpaired electrons. Hence spin value $\left(s = n \times \pm\frac{1}{2}\right)$ is zero.
(Q) Fe $\rightarrow [Ar]_{18}4s^2 3d^6$. In +2 O.S, it contains four unpaired electrons.
$\therefore$Magnetic moment $(\mu) = \sqrt{n(n+2)} = \sqrt{4(4+2)} = 2\sqrt{6}$ B.M
(R) Sc $\rightarrow [Ar]_{18}4s^2 3d^1$. In +3 O.S, it contains no d-electrons. Hence, it forms colourless compounds
(S) Ag $\rightarrow [Kr]_{36}5s^1 4d^{10}$. In +1 O.S, it contains no unpaired electrons. Hence it is diamagnetic.

SECTION – B

29. 4
Sol. Zn^{2+} occupy $\frac{1}{8}$th of tetrahedral voids and Fe^{3+} occupy $\frac{1}{2}$of octahedral voids.
So, $\frac{1}{2} : \frac{1}{8} = 4$

30. 6
Sol. $\Delta U = nC_v\Delta T$
or, $12 = \frac{4.48}{22.4} \times C_V \times 15$
or, $C_v = 4$
$C_P - C_V = R = 2 \Rightarrow C_P = C_V + 2 = 4 + 2 = 6$ cal

31. 317
Sol. P = $CrCl_3$
Q = $Cr(OH)_3$
R = $[Cr(OH)_4]^-$
S = Na_2CrO_4
T = $BaCrO_4$

x = 158.50 g/mol
$\therefore$ 2x = 317

32. 558

Sol. $Z_2O_3 + 3H_2 \longrightarrow 2Z + 3H_2O$

$$\left(\frac{0.1596}{2Z+48}\right) = \frac{6}{3\times 2}\times 10^{-3}$$

$$10^{-3} = \left(\frac{0.1596}{2Z+48}\right)$$

Z = 55.8 = x
$\therefore$ 10x = 558

33. 50

Sol. Dipole moment of the molecule having 100% ionic character is

$\mu_{ionic} = \left(4.8\times 10^{-10}\right)\times\left(1.25\times 10^{-8}\right) = 6 \times 10^{-18}$ esu-cm = 6 D

$\therefore$ % ionic character = $\frac{\mu_{obs}}{\mu_{ionic}}\times 100 = \frac{3}{6}\times 100 = 50\%$

34. 3

Sol. $M_{AcOH}V_{AcOH} = M_{NaOH}V_{NaOH}$

$M_{AcOH}\times 20 = 40\times 1.5$

$$M_{AcOH} = \frac{40\times 1.5}{20} = 3M$$

Mathematics

PART – III

SECTION – A

35. ABD

Sol. $DE: \left(1-x^2\right)\dfrac{dy}{dx}+xy=ax \Rightarrow \dfrac{dy}{dx}+\left(\dfrac{x}{1-x^2}\right)y=\dfrac{ax}{1-x^2}$.

This is a linear differential equation.

On solving, $x^2+\dfrac{(y-a)^2}{c^2}=1$ or $x^2-\dfrac{(y-a)^2}{c^2}=1$.

These are ellipse and hyperbola with centre (0, a).

36. AB

Sol. $z^4=iz \Rightarrow z=0$ or $z^3=i=\cos\dfrac{\pi}{2}+i\sin\dfrac{\pi}{2}$

$\Rightarrow z=0$ or $z=\cos\left(\dfrac{2k\pi}{3}+\dfrac{\pi}{6}\right)+i\sin\left(\dfrac{2k\pi}{3}+\dfrac{\pi}{6}\right)(k=0,1,2)$

$\Rightarrow z=0,\ -i,\dfrac{\sqrt{3}}{2}+\dfrac{i}{2},\ \dfrac{-\sqrt{3}}{2}+\dfrac{i}{2} \quad \Rightarrow z_1,z_2,z_3=-i,\pm\dfrac{\sqrt{3}}{2}+\dfrac{i}{2}$

37. BC

Sol. Let $D=\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_2 \end{bmatrix}=\text{dia}\left(d_1,d_2,d_3\right).D'=D$

For $A=\left[a_{ij}\right]_{3\times3}$, $AD\neq DA$ in general.

$D^{-1}=\text{dia}\left(\dfrac{1}{d_1},\dfrac{1}{d_2},\dfrac{1}{d_3}\right)$: It is not a scalar matrix.

38. B

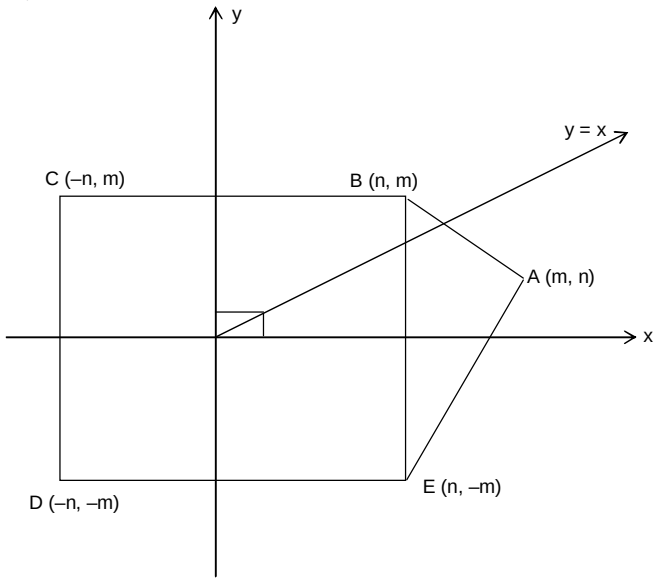
Sol. Reflection of A (m, n) about $y=x$ is B (n, m).

$\text{ar}(\text{Pentagon ABCDE})$

$=\text{ar}(\square\text{BCDE})+\text{ar}(\Delta\text{ABE})$

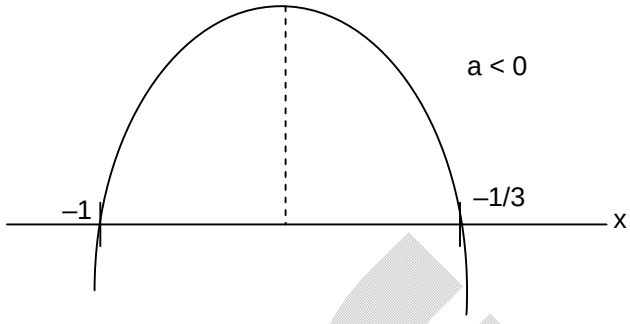
$=4mn+\dfrac{(2m)}{2}\times(m-n)$

$=m^2+3mn=m(m+3n)$.



39. B

Sol. Let $y = f(x) = ax^2 + bx + c$.



Then $f(-1) = f\left(-\frac{1}{3}\right) = 0$

So, $f(x) = 0$ has roots $x = -1$ and $-\frac{1}{3}$.

$f(x)$ has max at $x = -\frac{b}{2a}$

$a - b + c = 0$ .....(1)

$a - 3b + 9c = 0$ .....(2)

$\Rightarrow 8a - 6b = 0 \Rightarrow \frac{b}{a} = \frac{4}{3}$.

So, $-\frac{b}{2a} = -\frac{2}{3}$.

40. A

Sol. $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0 \Rightarrow (\vec{a}.\vec{c})\vec{a} - (\vec{a}.\vec{a})\vec{c} + \vec{b} = 0 \Rightarrow (\vec{a}.\vec{c})\vec{a} - |\vec{a}|^2 \vec{c} = -\vec{b}$

$\Rightarrow \left((\vec{a}.\vec{c})\vec{a} - \vec{c}\right)^2 = (-\vec{b})^2 \quad (|\vec{a}| = 1) \Rightarrow (\vec{a}.\vec{c})^2 |\vec{a}|^2 + |\vec{c}|^2 - 2(\vec{a}.\vec{c})(\vec{a}.\vec{c}) = |\vec{b}|^2 = 1$

$\Rightarrow (\vec{a}.\vec{c})^2 + 4 - 2(\vec{a}.\vec{c})^2 = 1 \quad (|\vec{c}| = 2) \Rightarrow (\vec{a}.\vec{c})^2 = 3 \Rightarrow |\vec{a}||\vec{c}|\cos\theta = \pm\sqrt{3} \Rightarrow \cos\theta = \pm\frac{\sqrt{3}}{2}$

acute angle $\theta = \frac{\pi}{6}$.

41. A

Sol. $P \equiv \left(n, \sqrt{1+n^2}\right)$. Then, $d_n = \frac{\left|\sqrt{n^2+1} - n\right|}{\sqrt{2}}$.

$\lim\limits_{n\to\infty}(n.d_n) = \lim\limits_{n\to\infty}\left(\frac{n\left(\sqrt{n^2+1} - n\right)}{\sqrt{2}}\right) = \lim\limits_{n\to\infty}\frac{n}{\left(\sqrt{n^2+1} + n\right)\sqrt{2}} = \frac{1}{2\sqrt{2}}$.

42. A

Sol. (P) $S = a + (a+d) + (a+2d) + ... + (a+98d) = \frac{99}{2}(2a + 98d) = 99(a + 49d)$ .....(1)

$2550 = a + (a+2d) + (a+4d) + ... + (a+98d)$ (Odd numbered terms)

$= \frac{50}{2}(2a + 98d) = 50(a + 49d) \Rightarrow a + 49d = \frac{2550}{50} = 51$

$S = 99(a + 49d) = 99 \times 51 = 5049$.

(Q) $f(n) = f(n-1) + n$

$f(2) = f(1) + 2 = 1 + 2$

$f(3) = f(2) + 3 = 1 + 2 + 3$

$f(4) = f(3) + 4 = 1 + 2 + 3 + 4$

and so on

So $f(100) = 1 + 2 + 3 + 4 + ... + 100 = \frac{100 \times 101}{2} = 5050$.

(R) $f(n) = \frac{\log 3}{\log 2} . \frac{\log 4}{\log 3} . \frac{\log 5}{\log 4} ..... \frac{\log n}{\log (n-1)} = \frac{\log n}{\log 2} = \log_2 (n)$

$\sum_{k=2}^{100} f\left(2^k\right) = \sum_{k=2}^{100} \log_2 \left(2^k\right) = \sum_{k=2}^{100} k = \frac{100 \times 101}{2} - 1 = 5049$.

(S) Area = $\pi\left(r_2^{\,2} - r_1^{\,2}\right) + \pi\left(r_4^{\,2} - r_3^{\,2}\right) + \pi\left(\pi_6^{\,2} - r_5^{\,2}\right) + ... + \pi\left(r_{100}^{\,2} - r_{99}^{\,2}\right)$

$= \pi\left(r_1 + r_2\right)\left(r_2 - r_1\right) + \pi\left(r_4 + r_3\right)\left(r_4 - r_3\right) + \pi\left(r_6 - r_5\right)\left(r_6 + r_5\right) + ... + \pi\left(r_{100} - r_{99}\right)\left(r_{100} + r_{99}\right)$

$= \pi\left(r_1 + r_2 + r_3 + r_4 + r_5 + ... + r_{100}\right)$

$= \pi\left(1 + 2 + 3 + 4 + 5 + ... + 100\right) = \pi\left(\frac{100 \times 101}{2}\right) = 5050\pi$

43. B

Sol. (P) $\left(1 + \omega + \omega^2 + ... + \omega^n\right)^m = \left(\frac{\omega^{n+1} - 1}{\omega - 1}\right)^m = \begin{cases} 0, \text{ if } n+1 = 3k, k \in \mathbb{Z} \\ 1, \quad \text{if } n+1 = 3k+1, \\ (\omega+1)^m, \text{ if } n+1 = 3k+2 \end{cases}$

$(\omega + 1)^m = \left(-\omega^2\right)^m = -1, 1, -\omega^2, \omega^2, \omega, -\omega$.

(Q) Non – real complex roots are in conjugate pairs.

$2\overline{\omega} = 2\omega^2$; $\overline{2 + 3\omega} = 2 + 3\omega^2$; $2 - \omega - \omega^2 = 3$

Roots are $2\omega, 2\omega^2, 2 + 3\omega$, $2 + 3\omega^2, 3$

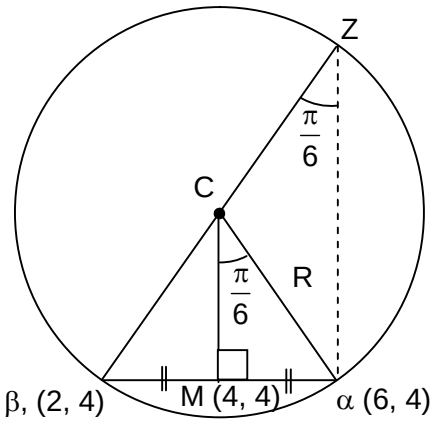
So, the least possible degree $= 2 + 2 + 1 = 5$.

(R) $\overline{z} = iz^2 \Rightarrow x - iy = i\left(x^2 - y^2 + 2ixy\right)$ (Let $z = x + iy$)

On solving, $z = 0, i, \frac{\sqrt{3}}{2} - \frac{i}{2}, -\frac{\sqrt{3}}{2} - \frac{i}{2}$.

(S) Refer to the figure.

$\sin\frac{\pi}{6} = \frac{2}{R} \Rightarrow R = 4$.



44. B

Sol. (P) $y = a\cos(x+b) - ce^x + d\sin x$

$= (a\cos b)\cos x - (a\sin b)\sin x - ce^x + d\sin x$

$= A\cos x + B\sin x - Ce^x$. (Order = 3)

(Q) $y = a\sin^2 x + b\cos^2 x + c\sin 2x + d\cos 2x + e\sin x$

$= \left(\frac{a+b}{2}\right) + \left(d + \frac{b}{2} - \frac{a}{2}\right)\cos 2x + c\sin 2x + e\sin x$

$= A + B\cos 2x + c\sin 2x + e\sin x$ (order = 4)

(R) $\left(1+y'^2\right)^3 = (y'')^2$. (degree = 2)

(S) Only one arbitrary constant. (Order = 1)

45. A

Sol. (P) $f(x) = \begin{cases} 0, & 1 < x \le 2 \\ 1-x, & 0 \le x < 1 \\ -\sin \pi x, & -1 \le x < 0 \\ 0, & x = 1 \end{cases}$.

So continuous at $x = 1$ but not differentiable at $x = 1$

(Q) $f'(0^-) = \lim\limits_{h\to 0^-} \frac{h^2 e^{\frac{1}{h}} - 0}{h} = \lim\limits_{h\to 0^-} h e^{\frac{1}{h}} = 0$.

(R) $g(x) = \frac{1}{1+\frac{2}{f(x)}} = \frac{x}{3x+2}$

f is not differentiable at $x = 0, -1$

So, g is not differentiable at $x = 0, -1$, and $\frac{-2}{3}$.

(S) $g(x) = \begin{cases} -1 & ; \ |x| < 1 \\ 0 & ; \ |x| = 1 \\ 1 & ; \ |x| > 1 \end{cases}$

$g(x)$ is not differentiable at $x = \pm 1$.

SECTION – B

46. 5

Sol. $1 + a = 4^{\frac{1}{401}}$. $b_n = \frac{1}{a}\left((1+a)^n - 1\right) = \frac{1}{a}\left[4^{\frac{n}{401}} - 1\right]$

$b_{2006} - b_{2005} = \frac{1}{a}\left(4^{\frac{2006}{401}} - 4^{\frac{2005}{401}}\right) = \frac{1}{a} 4^{\frac{2005}{401}} \cdot \left(4^{\frac{1}{401}} - 1\right)$

$= 4^{\frac{2005}{401}} = 4^5$

47. 5

Sol. $x_1 + x_2 + x_3 + x_4 + x_5 \le 20$

$\Rightarrow x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 20 \qquad (x_6 \ge 0,\ x_6 \in \mathbb{Z})$

$N(5) = {}^{20+6-1}C_{6-1} = {}^{25}C_5$

$N(4) = {}^{20+5-1}C_{5-1} = {}^{24}C_4$

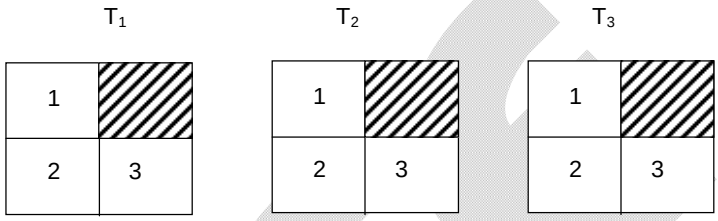
48. 8

Sol. $n(S) = {}^9C_7 \times 7!$

$n(A) = (7\times6\times5\times4\times3\times2)\times3$

Required probability

$= \dfrac{n(A)}{n(S)} = \dfrac{1}{12} = \dfrac{m}{n}.$



49. 4

Sol. Let $x = \tan\theta, 0 \le \theta \le \dfrac{\pi}{4}$. Then $u = \int_0^{\pi/4} \log(1+\tan\theta)\,d\theta$

$= \int_0^{\pi/4} \log\left(1+\tan\left(-\theta+\dfrac{\pi}{4}\right)\right) = \int_0^{\pi/4} \log\left(\dfrac{2}{1+\tan\theta}\right)d\theta = \int_0^{\pi/4} \log 2\,d\theta - u \Rightarrow u = \dfrac{\pi}{8}\log 2$

$v = \int_0^{\pi/2} \log(\sin 2x)\,dx = \dfrac{1}{2}\int_0^{\pi} \log\sin t\,dt \quad (\text{Let } t = 2x) = \int_0^{\pi/2} \log\sin t\,dt = \int_0^{\pi/2} \log\cos t\ dt$

$2v = \int_0^{\pi/2} (\log\sin t + \log\cos t)\,dt = \int_0^{\pi/2} \log(\sin t\cos t)\,dt$

$= \int_0^{\pi/2} (\log(\sin 2t) - \log 2)\,dt = v - \dfrac{\pi}{2}\log 2$

$\Rightarrow v = -\dfrac{\pi}{2}\log 2$

$\dfrac{-v}{u} = \dfrac{\frac{\pi}{2}\log 2}{\frac{\pi}{8}\log 2} = 4.$

50. 3

Sol. Let $\tan\theta = \dfrac{\pi}{\sqrt{2}}, \dfrac{\pi}{4} < \theta < \dfrac{\pi}{2}$. Then $\pi = \sqrt{2}\tan\theta$

$y = \dfrac{3}{2}\cos^{-1}\sqrt{\dfrac{2}{2+2\tan^2\theta}} + \dfrac{1}{4}\sin^{-1}\left(\dfrac{2\sqrt{2}\left(\sqrt{2}\tan\theta\right)}{2+2\tan^2\theta}\right) + \tan^{-1}(\cot\theta)$

$= \dfrac{3}{2}\theta + \dfrac{1}{4}(\pi - 2\theta) + \dfrac{\pi}{2} - \theta = \dfrac{3\pi}{4}.$

51. 80

Sol. Let $AB = p,\ AD = q$.

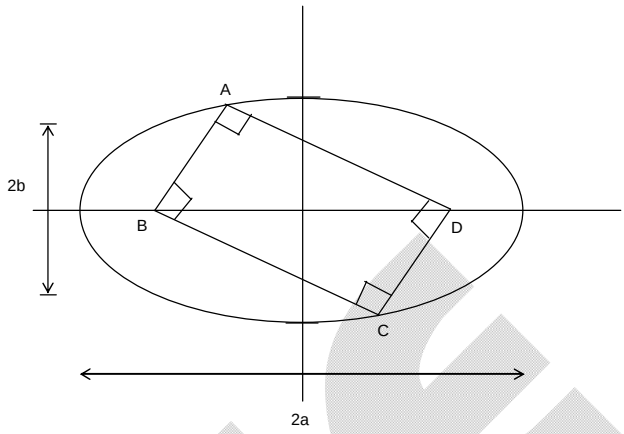
$AB + AD = 2a$

$\Rightarrow p + q = 2a$ (i)

$p^2 + q^2 = BD^2 = (2ae)^2 = 4a^2e^2$

$= 4(a^2 - b^2)$

$\Rightarrow (p+q)^2 - 2pq = 4(a^2 - b^2)$



$\Rightarrow (2a)^2 - 2(200) = 4(a^2 - b^2)$ ((Area $pq = 200$) and from (i)]

$\Rightarrow b = 10$

Area of ellipse $= \pi ab = 200\pi \Rightarrow ab = 200 \Rightarrow a = \dfrac{200}{10} = 20$

Perimeter of $\square ABCD = 2(p+q) = 2 \times 2a = 4 \times 20 = 80$.